

Produit : Electronic / Measuring devices

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INDIA

Product's definition

Category

In which category the product is classified ?

It is classified in the category of "

It is regulated by -

1.Battery Waste Management Rules, 2022

2. BIS -

(a) IS 8144 : 1997 - Multipurpose Dry Batteries Specifications

(b) IS 16270: 2023 - Secondary cells and batteries for solar photovoltaic application - General requirements and methods of test

(c)

Definition

What is the exact definition of the product ?

And if it exists, what is the interpretation of the authorities regarding it ?

The [Battery Waste Management Rules, 2022](#) define the term "Battery"

Rule 2(c):

"Battery" means new or refurbished cell and/or Battery and/or their component, including accumulator, which is any source of electrical energy generated by direct conversion of chemical energy and includes disposable primary and/or secondary battery;

However, there is no exact definition for accumulators under any specific regulation.

Rules for import, management and disposal of batteries are contained in the Batteries (Management and handling) Rules 2001.

IS 1652 (1991): Stationary cells and batteries, lead-acid type with plante positive plates [ETD 11: Secondary Cells and Batteries]

i.e Indian Standard-Stationary Cells And Batteries, Lead-Acid Type With Plante Positive Plates-Specification (Third Revision)

FOREWORD

This Indian Standard (Third Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Secondary Cells and Batteries Sectional Committee had been approved by the Electrotechnical Division Council.

This standard was first published in 1960, superseding IS 541 : 1954. The standard prescribed the dimensions, capacities and performance requirements of stationary cells and batteries of lead-acid type with plante positive plates. For preparation of this standard, assistance was derived from BS 440 : 1958 Stationary batteries (lead acid plante positive plates) for general electric purposes, issued by the British Standards Institution.

The first revision was brought out in 1972 to cover the storage cells with plastic containers and the plante cells of improved performance, thereby making available cells of a lower internal resistance and correspondingly higher cell output.

The second revision of this standard was brought out in 1984 to cover batteries with high discharge performance requirements as only the batteries with plante positive plates with high performance requirement are manufactured and used in the country. The batteries with low discharge performance requirements would be included later whenever there is a demand from the users and manufacturers of the batteries.

Third revision has been prepared to cover the latest practice in manufacture of stationary cells and batteries, and include additional requirements for overcharging cycle in endurance test.

For the purpose of deciding whether a particular requirement of this standard is complied with the firal value, observed or calculated, expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1 SCOPE

This standard specifies rated ampere-hour capacities, overall dimensions, performance requirements and tests for high discharge performance, stationary, lead-acid cells and batteries using plante type positive plates.

2 REFERENCES

The standards listed in Annex A are necessary ajuncts to this standard.

3 TERMINOLOGY

3.0 For the purpose of this standard, the following definitions in addition to those given in IS 1885 (Part 8) : 1986 shall apply.

3.1 High Discharge Performance Cells

Cells containing pure lead plate plates of lemelle type with pasted type negative plates.

3.2 Type Test

Tests carried out to prove conformity with the requirements of this standard. These are intended to prove the general quality and design of a given type of battery.

3.3 Acceptance Tests

Tests carried out on sample selected from a lot for the purpose of verifying the acceptability of the lot.

13.3.1 Lot

All batteries of the same type of design and rating manufactured by the same factory during the same period, using the same process and materials, offered for inspection at a time, shall constitute a lot.

IS 1885-15 (2008) : Electrotechnical Vocabulary, Part 15: Primary Cells and Batteries [Amalgamating IS 1885 (Part 8) : 1996] [ETD 1: Basic Electrotechnical Standards]

NATIONAL FOREWORD

This Indian Standard (Part 15) (Second Revision) which is identical with IEC 60050-482 : 2004 'International Electrotechnical Vocabulary – Part 482 : Primary and secondary cells and Batteries' issued by the recommendation of the Basic Electrotechnical Standards Sectional Committee and approval of the Electrotechnical Division Council.

This standard was first published in 1967. The first revision was published in 1983 and the second revision has been undertaken after amalgamating IS 1885 (Part 8) : 1986 'Electrotechnical vocabulary : Part 8 Secondary cells and batteries (first revision)' in order to bring it in line with the latest IEC publication 60050-484 ; 2004. After the publication of this standard IS 1885 (Part 8): 1986 shall be treated as withdrawn.

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

1. Wherever the words 'International Standard' appear referring to this standard. They should be read as 'Indian Standard'.
2. Comma (,) has been used as a decimal marker while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

In this adopted standard, reference appears to certain International standards for which Indian Standards also exist. The corresponding Indian Standards which are to be substituted in their respective places are given below along with their degree of equivalence for the editions indicated:

International Standard	Corresponding Indian Standard	Degree of Equivalence
IEC 600027-1 : 1992 Letter symbols to be used in electrical technology – Part 1: General	IS 3722 (Part 1) : 1983 Letter symbols and signs used in electrical technology: Part 1 General guidance on symbols and Subscripts (first revision)	Technically Equivalent
IEC 60050-151 : 2001 International Electrotechnical Vocabulary – Part 151: Electrical and magnetic devices	IS 1885 (Part 74) : 1993 Electrotechnical Vocabulary: Part 74 Electrical and magnetic devices	

The technical committee has reviewed the provisions of the following International Standard referred in this adopted standard and has decided that it is acceptable for use in conjunction with this standard:

International Standard	Title
IEC 60027-2 : 2005	Letter symbols to be used in electrical technology – Part 2: Telecommunications and electronics

For the purpose of deciding whether a particular requirement of this standard is complied with, with final value, observed or calculated, calculated, expressing the result of attest or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off the rounded off value should be same as that of that of the specified value in this standard.

1. Scope

This part of IEC 60050 gives the general terminology used in the field of primary and secondary cells and batteries, and reflects the technology, design, construction, performance and application employed.

This terminology is consistent with the terminology developed in other specialised parts of the IEC.

2. Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60027-1:1992	Letter symbols to be used in electrical technology – Part 1: General + Amendment 1:1997
IEC 60027-2:2000	Letter symbols to be used in electrical technology – Part 2: Telecommunications and electronics
IEC 60050-151: 2001,	International Electrotechnical Vocabulary – Part 151: Electrical and magnetic devices

3. Terms and definitions

Section 482-01 – Fundamental concepts

(please refer to Pdf File attached)

Section 482-02 – Parts, components, accessories, shape

(please refer to Pdf File attached)

Section 482-03 – Electrical properties and operation

(please refer to Pdf File attached)

Section 482 -04 – Terms typically used for primary cells and batteries

(please refer to Pdf File attached)

Section 482-05 – Terms typically used for secondary cells and batteries

(please refer to Pdf File attached)

Labelling

Language

In which language must the information on the product or on the packaging be indicated? If English is sufficient please indicate NA.

Note: Generally All the particulars and Labelling are made in English Language.

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

9. Manner in which declaration shall be made

4) The particulars of the declarations required to be specified under this rule on a package shall either be in Hindi in Devanagiri script or in English; Provided...

Mandatory labelling on product

What are the mandatory labelling on the product ?

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7 MARRKING

The following information shall be indelibly and durably marked on the outside of the cell:

- a) indicating the source of manufacturer,
- b) Ampere-hour capacity at IO-hour rate:
- c) Upper and lower electrolyte level in case of transparent containers;
- d) Year of manufacture, and
- e) Country of origin.

TABLE 1 Capacities and Maximum overall Dimension at Cells Boxes (Clauses 6.1

Mandatory labelling on packaging

What are the mandatory labelling on the packaging ?

A. The Battery Waste Management Rules, 2022 (Amendment of February 2025) -

“(ib) producers may fulfill the requirements of clause (ia), subject to providing the information in writing to the Central Pollution Control Board,

— A. print a barcode or Quick Response code containing the Extended Producer Responsibility registration number on —

(a) battery or battery pack; or

(b) equipment having battery or battery pack; or

(c) packaging of battery or battery pack; or

(d) packaging of the equipment having battery or battery pack; or

(e) bulk packaging of batteries or battery packs, not for retail sale;

B. print the Extended Producer Registration number on the product information brochure; Provided that the Central Pollution Control Board shall publish a consolidated list of such producers, who have provided the information to it on the centralised online portal and update their details every quarter.”.

B. The Legal Metrology (Packaged Commodities) Rules, 2011

Every packaged product should have the following declarations in the form of an MRP Label affixed on it -

1. Maximum Retail Price
2. Month and Year of Import/Manufacturer
3. Name and Address of the Importer/Manufacturer/Packer
4. Generic Name of the Product (Nature of Product)
5. Size of the product (ml, l, cm, m, gms, kg)
6. Quantity (number of units in the package)
7. Price Per Unit
8. Country of Origin
9. Best Before / Use By Date (in case of perishable products)

The above declarations is updated on the MRP tool developed for India. Suppliers across the world need to feed in the model code / item code and the sticker auto generates and can be printed. The printed sticker should be affixed on the package ensuring that no other declarations in English are covered, and there is no fold / restriction to the MRP sticker. It should be on a flat surface.

NOTE:

For Electronic Goods Manufactured/Imported after the 15th of July 2022 - the package of such product shall, for a period of one year from such date, declare the name of the manufacturer or packer or importer, as the case may be, on the package

itself and such declaration shall also inform the consumers to scan the QR code for the address and other related information, in case such information is declared through the QR Code and not declared on the package itself.

Mandatory labelling on the instructions manual

What are the mandatory labelling on the instructions manual (if one is required) ?
In which format, do we have to deliver the instruction manual to our customers ?

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9 MANUAL AND INSTRUCTIONS

9.1 The manufacturer shall supply one copy of instruction manual for initial treatment and routine maintenance during service, with every batch of batteries.

9.2 The following information shall be provided on the instructions cards:

- a) Designation of cell or battery (see 4.2);
- b) Ampere-hour capacity;
- c) Nominal voltage;
- d) Manufacturer's instructions for filling, initial charging;
- e) Normal and finishing charging rates; and
- f) Maintenance instruction.

Warnings - General ones

Are general warnings required ? If yes, which ones ?

Not Applicable

Warnings - Specific ones

Are specific warnings required ? If yes which ones ?

Not Applicable

Target Age

Is there any target age applicable? If so, does it have to be indicated and in which form?

Not Applicable

Sanctions / Fines

What are the sanctions / fines in case of non conformity ?

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

20. Action to be taken on completion of inspection of packages at the premises of the manufacturer or the packer:- (1) If it appears from the report referred to in sub-rule (3) of rule 19 that,-

(a) the statistical average of the net quantity contained in the packages drawn as samples under that rule is lesser than the quantity declared on the packages or on the labels affixed thereto, or any such package shows an error in deficiency greater than the maximum permissible error, or

(b) any such package does not bear thereon or on label affixed thereto the declarations to be made under these rules, the Director, Controller or any Legal Metrology Officer shall take action as given below, namely:-

(i) seize the packages drawn by him as samples and shall take adequate steps for the safe custody of seized packages until they are produced in the appropriate court as evidence;

(ii) based on the evidence initiate action for violations of the provisions of the Act and these rules:

Provided that no such action shall be taken if fresh tests are carried out under sub-rule (4) of rule 19, but if after such fresh tests any error or omission as is referred to in this sub-rule is detected, the Director, Controller or any Legal Metrology Officer shall take appropriate action as specified in this sub-rule in accordance with the provisions of the Act against the manufacturer or, as the case may be, the packer.

(3) The disposal of the seized packages shall be made in accordance with the provisions of the Code of Criminal Procedure, 1973 (2 of 1974).

Chapter VII GENERAL

32. Penalty for contravention of Rules.-(1) Whoever contravenes the provisions of rules 27 to 31, he shall be punished with fine of four thousand rupees.

(2) Whoever contravenes any other provision of these rules, for the contravention of which no punishment has been provided either in the Act or in the rules, he shall be punished with fine of two thousand rupees.

Comments

Comments

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

18. Provisions relating to wholesale dealer and retail dealers: (Refer to authority attached)

19. Inspection of quantity and error in packages at the premises of the manufacturer or packer:- (Refer to authority attached)

21. Inspection of quantity and error in packages at the premises of the wholesale dealer or retail dealer :- (Refer to authority attached)

23. Deceptive packages to be repacked or in default to be seized: (Refer to authority attached)

Chapter IV EXEMPTIONS

26. Exemption in respect of certain packages: - Nothing contained in these rules shall apply to any package containing a commodity if-

(a) the net weight or measurement of the commodity is ten gram or ten milliliter or less, if sold by weight or measurement.(**)

Chapter VI REGISTRATION OF MANUFACTURERS, PACKERS AND IMPORTERS

27. Registration of manufacturers, Packers and importers :(Refer to authority attached)

28. Registration of shorter address permissible :.. (Refer to authority attached)

29. Registration of manufacturers and packers, etc.: .. (Refer to authority attached)

Chapter VII GENERAL

31. (1) Any advertisement mentioning the retail ale price of the pre-packaged commodity shall contain a declaration as to the net quantity or number of the commodity contained in the package.

(2) The font size of the net quantity in the advertisement shall be same as that of retail sale price.

30. Compilation of lists of manufacturers or packers and their circulation...

33. Repeal and savings ...

Environment

Recycling - Eco taxes

Which regulations concerning ecodesign and recycling apply to this product? Are there any ecotaxes applicable to this category of product?

Section 4 - the international 'recycling' sign must be featured on the batteries.

Labelling

What are the mandatory labellings to put on the product or on the packaging regarding the environmental rules ?

Technical rules / Standards

General safety requirements

What are the general safety requirements ? Does a risk assessment have to be conducted ? If yes, does it concern specific risk ?

IS 1652 (1991): Stationary cells and batteries, lead-acid type with plante positive plates [ETD 11: Secondary Cells and Batteries]

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11 GENERAL REQUIREMENT FOR TEST

11.1 Temperature for Test

The temperature of electrolyte during test dis- charge shall be within the limits of 160 degree C and 38 degree C.

11.2 Specific Gravity

For the purpose of test requirements, the specific gravity of electrolyte of fully charged cell shall be 1.200 ± 0.005 corrected to 27 degree C. Temperature correction for hydrometer readings of specific gravity shall be made as follows:

1. For each 1 Degree C, above 27 degree C, 0.000 7 to be added to the observed reading and
2. For each 1 degree C, below 27 degree C, 0.000 7 to be subtracted from the observed reading.

11.3 Test Equipment

The voltmeter, ammeters, thermometers and hydrometers used for the tests shall comply with the requirements of 10.2 of IS 8320 : 1982.

11.4 Classification of Tests

11.4.1 Type Tests

The following shall constitute type tests and shall be carried out in the sequence given below:

1. Verification of constructional requirements (5),
2. Verification of marking (7),
3. Verification of dimensions (6.1),
4. Test for capacity (11.6) - Test for voltage during discharge (11.11),
5. Ampere-hour and watt-hour efficiency tests (11.10)
6. Test for retention of charge (11.8), and T
7. Endurance test (11.9).

11.4.2 Type test shall be carried out on three (3) cells. Two (2) cells shall undergo all tests excepts endurance test while the other would undergo endurance test preceded by a capacity test.

For capacities above 500 Ah, however a test cell with suitable number of positive plates cases alongwith other components to be fabricated in a suitable sized container such that capacity does not exceed 500 Ah. All other tests would be carried out on two (2) cells of respective capacity.

11.4.3 Acceptance Tests

The following shall constitute the acceptance tests.

1. Verification of marking (7.1),
2. Verification of dimensions (6.1),
3. Test for capacity (11.6), and
4. Test for voltage during discharge (11.11)

11.4.3.1 Acceptance tests shall, normally, be carried out, at the discretion of the purchaser, on each cell after installation at site. The date and place of testing shall be subject to agreement between the purchaser and the supplier.

11.4.3.2 The sampling scheme and criteria for acceptance for cells up to and including 600 Ah capacity shall be in accordance with 11.1.4 of IS 8320 : 1982. For capacity more than 600 Ah, it shall be as agreed between the purchaser and the manufacturer.

11.5 First Charge

The cell of battery, if received in the dry uncharged condition, shall be filled with the electrolyte and charged in accordance with the manufacturer's instructions.

11.6 Tests for Capacity

11.6.1 After standing on open circuit for not less than 12 hours and not more than 24 hours from completion of a full charge, the cell or battery shall be discharged through a suitable variable resistance at constant current of $I = 0.1 C_{10} A$. (refer to pdf attached). The discharge shall be stopped when the closed circuit voltage across the battery falls to $1.85 \times nV$, where n is the number of cells.

11.6.1.1 The rated capacity shall be reached within 3 discharge subsequent to the initial charge. Once the rated capacity has been met, further discharges for capacity shall be discontinued.

11.6.2 Requirement tests

The period of discharge shall not be less than $10 + 0.09 (t - 27)$ hours nor exceed $12 - 0.09 (t - 27)$ hours, t being the test temperature in degrees Celsius.

11.7 Alternate Rated Test Discharge

Test for capacity may, by agreement between the purchaser and the supplier, be carried out at rates other than the 10-hour rate. In such cases, the 5-hour rate is recommended, capacities at various rates of discharge and corresponding final voltages are given in Table 2. For the purpose of acceptance, the capacity test shall be carried out at one rate only.

(Please refer to page 9 in PDF file attached for Table 2)

11.8 Loss of Capacity on Storage

11.8.1 This test shall be carried out on the two cells which have successfully passed the capacity test in accordance with 11.6

11.8.2 The cells shall be fully recharged at the current specified by the manufacturer and shall then be submitted to two consecutive capacity tests in accordance with 11.6, the value of the initial capacity C being calculated as the mean of the two results thus obtained.

11.8.3 After a complete recharge and the cleaning of electrolyte from its surface the cells shall be left on open circuit for a period of 28 days without disturbance at a temperature of 27 ± 2 Degree C.

11.8.4 After the storage for 28 days, the cell shall be discharged in accordance with 11.6. The value of capacity measured after storage is denoted by C'.

11.8.5 The loss of capacity S expressed as a percentage is calculated from the following formula:

[Please refer to PDF file attached for formula]

11.8.6 Requirement

The loss in capacity thus measured shall not exceed 15 percent.

11.9 Endurance Test

11.9.1 The test shall be made on cell, which has been subjected to only capacity test. No further tests shall be made on this sample after this test.

11.9.2 The test shall be conducted on a complete cell up to a capacity 500 Ah. For those cells above this rating, test cells with suitable number of positive plates (of the type used in that particular cell alongwith negative plates and separators and other accessories) such that the cell capacity does not exceed 500 .4h, shall be built in a suitable sized container.

11.9.3 One cell shall be subjected to an over- charging current of 0.1×0.10 Amperes (refer to PDF file here) at a test temperature of 40 ± 3 degree C for total periods of 2 000 hours as given in 11.9.4.

11.9.4 Overcharging Periods

- a) 2-cycles of 300 h charging followed by test discharge,
- b) 3-cycles of 200 h charging followed by test discharge, and
- c) 8 cycles of 100 h charging followed by test discharge.

11.9.5 Throughout these periods of charge the cells shall be immersed in a tank of water the temperature of which shall be maintained at 40 ± 3 degree C. The cells shall be so immersed that the top of each cell is 25 mm above the water level in the tank. If the two cells are placed in the same tank, a distance between a cell and the sides of the tank shall be at least 25 mm.

11.9.6 At the end of each period of charging (a), (b), (c), as specified in 11.9.4 the cells shall be subjected without redrawing to a test discharge (C10) to an end voltage 1.85 V in accordance with 11.6.

11.9.7 When the test discharge is completed, the cells shall be immediately subjected to the next period of continuous charge without any recharging.

11.9.8 Requirement

The cell shall give not less than 9 h of discharge of the final test discharge after the last period of charge.

11.10 Ampere-Hour and Watt-Hour Efficiency Test

11.10.1 Ampere-Hour Efficiency

A fully charged cell shall be discharged at $I = 0.1 \times C_{10}$ A (please refer to PDF file attached) to an end voltage of 1.85 volts, careful calculations being made of the exact number of ampere-hours delivered. On recharge the same number of ampere-hours are put back at the same current. A second discharge shall then be made to the same out-off voltage as before. The efficiency of the cell is then calculated as the ratio of the ampere-hour delivered during the second discharge to the ampere-hour put in on the charge.

11.10.1.1 Requirement

The ampere-hour efficiency when calculated as described in 11.10.1 shall be not less than 10 percent

11.10.2 Watt-Hour Efficiency

The watt-hour efficiency shall be calculated by multiplying the ampere-hour efficiency by the ratio of average discharge and recharge voltage. The values of discharge and recharges voltage shall be calculated from the log sheets for ampere-hour efficiency.

11.10.2.1 Requirement

The watt-hour efficiency when calculated as described in 11.10.2 shall be not less than 75 percent.

11.11 Test for Voltages During Discharge

The cell need not be discharged specially for this test. For the purposes of this test the voltages shall be obtained from the log sheets for capacity test in which the cell meets the rated capacity.

11.11.1 Requirement

The closed-circuit voltage of cell shall not be below the values given in table 2.

Duration After Commencement of Discharge

Closed-circuit Voltage at Cell Terminals

(1)

(2)

a) Six minutes after putting on load

1.99 Min

b) 7 hours

1.92 Min

c) 10 hours

1.85Min

Mandatory or voluntary standards

Are there some mandatory or voluntary standards applicable ? If yes, which ones ?

Please indicate the reference of the applicable standards

IS 1652 (1991): Stationary cells and batteries, lead-acid type with planté positive plates [ETD 11: Secondary Cells and Batteries]

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4 RATING AND DESIGNATION

4.1 Ampere-Hour Rating

The rating assigned to the cell shall be the capacity expressed in ampere-hours (after, correction to 27 Degree C) stated by the manufacturer to be obtainable when the cell is discharged at the 10 hour rate (C 10) to a final voltage of 1.85 volts.

4.2 Designation of Cells

Cells shall be designated by the letter 'P' for planté followed by the standard rating of the cells, a letter indicating the type of cell container a hyphen and the letters HDP for high discharge performance.

The following will be used to indicate container types:

G = glass

H = hard rubber

P = plastics

W = lead-lined wood

f = fibre reinforced plastics (FRP)

Examples:

1. P 40 P-HDP designates a high discharge performance planté type stationary cell of 40 Ah capacity in a plastic container.
2. P 2000 W-HDP designates a high discharge performance cell of 2 000 Ah capacity in a lead-lined wood box.

5 CONSTRUCTIONAL REQUIREMENTS

5.1 Containers

The containers shall be made of hard rubber, glass, lead-lined wood, plastics or fibre re- inforced plastics (FRP).

5.1.1 Hard rubber and plastics containers shall conform to IS 1146 : 1981.

5.1.2 Glass containers shall be sufficiently robust, transparent and free from flaws.

5.2 Cell Lids

Lids used with sealed or closed type cells shall be of glass, plastics or ebonite and shall be provided with a vent plug(s). Terminal posts shall be suitably sealed at the lid by means of rubber grommets, sealing compound or other suitable devices, to prevent acid spray.

5.3 Venting Device

The venting device shall be of antispash type with more than one exit hole and shall allow gases to escape freely but shall effectively prevent acid particles or spray from coming out. For capacities 120 Ah and above assembled in opaque container, there shall be two vent holes, one serving as a guide for acid level, indicator for checking of the electrolyte level and the other to permit drawing of electrolyte samples, servicing, checking of specific gravity etc.

5.4 Sealing Compound

The sealing compound when used in sealed or closed cells, if bitumen based, shall conform to IS 3116 : 1965.

5.5 Plates

5.5.1 Positive Plate

The positive plate shall be of the pure lead lamelle type with plate formation.

5.5.2 Negative Plate

The plates shall be of good construction and of good workmanship.

5.6 Separators

The separators shall conform to IS 6071 : 1986.

5.7 Terminal Posts

Positive and negative terminal posts (where provided) of cells shall be clearly and unmistakably identifiable. The positive terminal shall be marked with red colour in addition to '+' marking.

5.8 Fasteners

Bolts, nuts and washers for connecting the cells shall be effectively lead-coated to prevent corrosion.

5.9 Connectors

Where it is not possible to bolt the cell terminals directly to assemble a battery, separate lead-coated copper or aluminium connectors of suitable size shall be provided to join the cells. In such cases, the coating shall be adequate and tenacious. In some cases, it may be necessary to connect individual cells in parallel as specified by the purchaser. In such cases, suitable connectors shall be used.

5.10 Electrolyte

The sulphuric acid and water used for preparing electrolyte for the cells shall conform to IS 266 : 1977 and IS 1069 : 1964.

5.11 Spray Arrestor

Open cells shall be provided with spray arrestors of adequate area over the plates. These may be of glass sheet at least 3 mm thick or suitable plastics sheet and shall be adequately supported.

6 CAPACITIES AND DIMENSIONS

The capacities and maximum external dimensions of cell boxes shall conform to Table 1. The voltage of each cell shall be 2 V.

10 CONDITIONS OF SUPPLY

10.1 To facilitate procurement of correct type of stationary cells, it is recommended that the user should furnish information regarding his requirements as given in Annex B, at the time of enquiry or order similarly, the manufacturer should furnish the information given in Annex C.

10.2 Other conditions of supply shall be subject to agreement between the purchaser and the supplier.

ANNEX A (refer to PDF file attached)

ANNEX B (refer to PDF file attached)

ANNEX C (refer to PDF file attached)

IS 1885-15 (2008) : Electrotechnical Vocabulary, Part 15: Primary Cells and Batteries [Amalgamating IS 1885 (Part 8) : 1996] [ETD 1: Basic Electrotechnical Standards]

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International Standard	Title
IEC 60027-2 : 2005	Letter symbols to be used in electrical technology – Part 2: Telecommunications and electronics

For the purpose of deciding whether a particular requirement of this standard is complied with, with final value, observed or calculated, expressing the result of attest or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off the rounded off value should be same as that of that of the specified value in this standard.

1. Scope

This part of IEC 60050 gives the general terminology used in the field of primary and secondary celles and batteries, and reflects the technology, design, construction, performance and application employed.

This terminology is consistent with the terminology developed in other specialised parts of the IEV.

2. Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60027-1:1992	Letter symbols to be used in electrical technology – Part 1: General + Amendment 1:1997
IEC 60027-2:2000	Letter symbols to be used in electrical technology – Part 2: Telecommunications and electronics
IEC 60050-151: 2001,	Internationals Electrotechnical Vocabulary – Part 151: Electrical and magnetic devices

3. Terms and definitions

Section 482-01 – Fundamental concepts

(please refer to Pdf File attached)

Section 482-02 – Parties, components, accessories, shape

(please refer to Pdf File attached)

Section 482-03 – Electrical properties and operation

(please refer to Pdf File attached)

Section 482 -04 – Terms typically used for primary cells and batteries

(please refer to Pdf File attached)

Section 482-05 – Terms typically used for secondary cells and batteries

(please refer to Pdf File attached)

Physical and mechanicals properties

What are the physical and mechanicals properties required ?

Refer to question of “Mandatory Standards” in the Technical Section.

Electrical aspects

What are requirements regarding electrical aspects ?

Refer to question of “Mandatory Standards” in the Technical Section.

Inflammability

Are there any specific requirements to ensure inflammability?

Not Applicable

Hygiene

What are the rules regarding hygiene ?

Not Applicable

Radioactivity

What are the rules regarding radioactivity ?

Not Applicable

Radiofrequency (2G,3G,4G,5G, Bluetooth, Ant+, GPS, Wifi...)

What are the rules regarding the radiofrequency?

Products making noise

What are the requirements regarding products making noise ?

Not Applicable

Chemicals substances

What are the requirements regarding chemicals substances ? Which ones are forbidden ? Limited ? Do we have to clearly indicate all substances that are in the product, or is the indication of major substances enough?"

Refer to question of "Mandatory Standards" in the Technical Section.

Packaging requirements

Can you please indicate if there are some requirements regarding the packaging of our products (mainly they are in paper, carton, plastic, textile) ? If yes, please explain them.

Not Applicable

Laboratory

In which laboratory can we perform the testing of the product according to the standard?

Please refer to following pdf files uploaded :

Group 1 List of BIS Recognised Laboratories as on 13-11-2019

Group 2 Lab of National Repute and Eminence, Facilities

Sanctions / Fines

What are the sanctions / fines in case of non conformity ?

Not Applicable

Comments

Comments

Certifications / Registrations

Certifications or registrations

Are there some certifications or registrations applicable to the product ?

1. Multipurpose Dry Batteries as per IS 8144 falls under the Scheme 1 (ISI Mark Scheme) under BIS. Therefore, the products under this category require the ISI mark.

2. Secondary Cells and Batteries containing Alkaline or other non-acid Electrolytes-Safety Requirements for Portable sealed secondary cells, and for Batteries made from them for use in portable applications as per IS 16046 falls under Scheme II (Registration Scheme) under BIS.

3. Secondary Cells and Batteries for Solar Photovoltaic Application General-Requirements and Methods of Test as per IS 16270 falls under Scheme II (Registration Scheme) under BIS.

Laboratory

In case of mandatory certification/ registration that we must follow please precise the name and address of any laboratories where we can conduct them

Comments

Comments

Conformity documents

Documents

What are the conformity documents ? Can you please list them ?

Not Applicable

For ISI Certificate, refer to comments sections of "Certifications/Registrations".

Format

Does it exist a specific format for the conformity documents ? If yes, can you please provide us an example in english ?

No Particular Format

Comments

Comments

Importation rules

General rules

What are the general rules in order to import the product ?

As per the [Battery Waste Management Rules, 2022](#)-

Importer of batteries is also considered as a "Producer"

Rule 3 (u) (iii) " Producers" shall mean - Importers are considered "Producers" under these rules and must register with the Central Pollution Control Board (CPCB).

- Registration:
 - Mandatory CPCB registration through the centralized EPR portal.
 - No entity (importer, recycler, refurbisher) shall operate without registration.
- Labelling Requirements:
 - Batteries must be labeled with appropriate information, including the "crossed-out wheeled bin" symbol and chemical content (e.g., Pb, Cd, Hg), to indicate hazardous nature.
- Prohibition:
 - Sale or import of batteries that do not conform to BIS standards is prohibited.
 - Prohibited from disposing of batteries in landfills or through incineration.

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Generally, Importers are required to register with the DGFT to obtain an Importer Exporter Code Number (IEC) issued against their Permanent Account Number (PAN), before engaging in EXIM activities. After an IEC has been obtained, the source of items for import must be identified and declared.

IMPORT PROCEDURE

Step1 : File Bill of Entry with Business Identification Number (BIN)

Step 2: Determine rate of duty for clearance from warehouse

Step 3: File requisite documents with customs department

Step 4: Submit import report/ manifest

Step 5: Receive permission to import goods

Documents

What are the required documents ?

Mandatory Documents:

1. Importer Exporter Code issued from the DGFT
2. Bill of Entry
3. Examination Order generated by the EDI system of Customs

Optional Documents depending upon the type of Product being Imported:

1. Certificate of Origin issued by Authorized Person/ Agency at the place of manufacturing/ processing etc. of the food consignment. Certificate of Origin shall contain information on Country of Origin etc. Hence if the consignor is from a different country;
2. Stuffing list, Packing List;
3. Commercial invoice as mentioned in the Bill of Entry (BoE);
4. Bill of Lading as mentioned in the Bill of Entry (BoE) for sea consignment;
5. Air Way Bill as mentioned in the Bill of Entry (BoE) for air consignment;

Batteries (Management and Handling) Rules 2001 Section 6

Section 6 - Customs clearance of imports is contingent upon

1. valid registration with the Reserve Bank of India (with Importer's Code Number); (ij)one time registration with the Ministry of Environment and Forests or an agency designated by it in Form-11;
2. undertaking in Form-111; and (iv)a copy of the latest half-yearly return in Form-IV

Importers of lead batteries must also submit an undertaking under Form III for collection of used batteries.

Time period

In which time period may we obtain the right to import ?

It generally takes 45 to 60 days for completing the overall procedure from import clearance to Product reaching store shelf.

Administration

What is the administration in charge of importations ? Name and address ?

- Directorate General of Foreign Trade
- Central Board of Excise and Customs

Batteries (Management and Handling) Rules 2001

Section 12 - The prescribed authority for ensuring compliance of the provisions of the Rules, is the (respective) State Pollution Control Board. Additionally an annual compliance status report must be submitted to the Central Pollution Control Board by 30th April of every year.

Comments

Comments

Inspection of the product

Inspection method

Does it exist an inspection procedure applicable to the product ? If yes, which administration is in charge of ? Can you please describe the procedure ?

IS 1652 (1991): Stationary cells and batteries, lead-acid type with positive plates [ETD 11: Secondary Cells and Batteries]

i.e Indian Standard-Stationary Cells And Batteries, Lead-Acid Type With Positive Plates-Specification (Third Revision)

11 GENERAL REQUIREMENT FOR TEST

11.1 Temperature for Test

The temperature of electrolyte during test discharge shall be within the limits of 160 degree C and 38 degree C.

11.2 Specific Gravity

For the purpose of test requirements, the specific gravity of electrolyte of fully charged cell shall be 1.200 ± 0.005 corrected to 27 degree C. Temperature correction for hydrometer readings of specific gravity shall be made as follows:

1. For each 1 Degree C, above 27 degree C, 0.000 7 to be added to the observed reading and
2. For each 1 degree C, below 27 degree C, 0.000 7 to be subtracted from the observed reading.

11.3 Test Equipment

The voltmeter, ammeters, thermometers and hydrometers used for the tests shall comply with the requirements of 10.2 of IS 8320 : 1982.

11.4 Classification of Tests

11.4.1 Type Tests

The following shall constitute type tests and shall be carried out in the sequence given below:

1. Verification of constructional requirements (5),
2. Verification of marking (7),
3. Verification of dimensions (6.1),
4. Test for capacity (11.6) - Test for voltage during discharge (11.11),
5. Ampere-hour and watt-hour efficiency tests (11.10)
6. Test for retention of charge (11.8), and T
7. Endurance test (11.9).

11.4.2 Type test shall be carried out on three (3) cells. Two (2) cells shall undergo all tests excepts endurance test while the other would undergo endurance test preceded by a capacity test.

For capacities above 500 Ah, however a test cell with suitable number of positive plates cases alongwith other components to be fabricated in a suitable sized container such that capacity does not exceed 500 Ah. All other tests would be carried out on two (2) cells of respective capacity.

11.4.3 Acceptance Tests

The following shall constitute the acceptance tests.

1. Verification of marking (7.1),
2. Verification of dimensions (6.1),
3. Test for capacity (11.6), and
4. Test for voltage during discharge (11.11)

11.4.3.1 Acceptance tests shall, normally, be carried out, at the discretion of the purchaser, on each cell after installation at site. The date and place of testing shall be subject to agreement between the purchaser and the supplier.

11.4.3.2 The sampling scheme and criteria for acceptance for cells up to and including 600 Ah capacity shall be in accordance with 11.1.4 of IS 8320 : 1982. For capacity more than 600 Ah, it shall be as agreed between the purchaser and the manufacturer.

11.5 First Charge

The cell of battery, if received in the dry uncharged condition, shall be filled with the electrolyte and charged in accordance with the manufacturer's instructions.

11.6 Tests for Capacity

11.6.1 After standing on open circuit for not less than 12 hours and not more than 24 hours from completion of a full charge, the cell or battery shall be discharged through a suitable variable resistance at constant current of $I = 0.1 C_{10} A$. (refer to pdf attached). The discharge shall be stopped when the closed circuit voltage across the battery falls to $1.85 \times nV$, where n is the number of cells.

11.6.1.1 The rated capacity shall be reached within 3 discharge subsequent to the initial charge. Once the rated capacity has been met, further discharges for capacity shall be discontinued.

11.6.2 Requirement tests

The period of discharge shall not be less than $10 + 0.09 (t - 27)$ hours nor exceed $12 - 0.09 (t - 27)$ hours, t being the test temperature in degrees Celsius.

11.7 Alternate Rated Test Discharge

Test for capacity may, by agreement between the purchaser and the supplier, be carried out at rates other than the 10-hour rate. In such cases, the 5-hour rate is recommended, capacities at various rates of discharge and corresponding final voltages are given in Table 2. For the purpose of acceptance, the capacity test shall be carried out at one rate only.

(Please refer to page 9 in PDF file attached for Table 2)

11.8 Loss of Capacity on Storage

11.8.1 This test shall be carried out on the two cells which have successfully passed the capacity test in accordance with 11.6

11.8.2 The cells shall be fully recharged at the current specified by the manufacturer and shall then be submitted to two consecutive capacity tests in accordance with 11.6, the value of the initial capacity C being calculated as the mean of the two results thus obtained.

11.8.3 After a complete recharge and the cleaning of electrolyte from its surface the cells shall be left on open circuit for a period of 28 days without disturbance at a temperature of 27 ± 2 Degree C.

11.8.4 After the storage for 28 days, the cell shall be discharged in accordance with 11.6. The value of capacity measured after storage is denoted by C'.

11.8.5 The loss of capacity S expressed as a percentage is calculated from the following formula:

[Please refer to PDF file attached for formula]

11.8.6 Requirement

The loss in capacity thus measured shall not exceed 15 percent.

11.9 Endurance Test

11.9.1 The test shall be made on cell, which has been subjected to only capacity test. No further tests shall be made on this sample after this test.

11.9.2 The test shall be conducted on a complete cell up to a capacity 500 Ah. For those cells above this rating, test cells with suitable number of positive plates (of the type used in that particular cell alongwith negative plates and separators and other accessories) such that the cell capacity does not exceed 500 .4h, shall be built in a suitable sized container.

11.9.3 One cell shall be subjected to an over- charging current of $0.1 \times C_{10}$ Amperes (refer to PDF file here) at a test temperature of 40 ± 3 degree C for total periods of 2 000 hours as given in 11.9.4.

11.9.4 Overcharging Periods

- a) 2-cycles of 300 h charging followed by test discharge,
- b) 3-cycles of 200 h charging followed by test discharge, and
- c) 8 cycles of 100 h charging followed by test discharge.

11.9.5 Throughout these periods of charge the cells shall be immersed in a tank of water the temperature of which shall be maintained at 40 ± 3 degree C. The cells shall be so immersed that the top of each cell is 25 mm above the water level in the tank. If the two cells are placed in the same tank, a distance between a cell and the sides of the tank shall be at least 25 mm.

11.9.6 At the end of each period of charging (a), (b), (c), as specified in 11.9.4 the cells shall be subjected without redrawing to a test discharge (C_{10}) to an end voltage 1.85 V in accordance with 11.6.

11.9.7 When the test discharge is completed, the cells shall be immediately subjected to the next period of continuous charge without any recharging.

11.9.8 Requirement

The cell shall give not less than 9 h of discharge of the final test discharge after the last period of charge.

11.10 Ampere-Hour and Watt-Hour Efficiency Test

11.10.1 Ampere-Hour Efficiency

A fully charged cell shall be discharged at $I = 0.1 \times C_{10}$ A (please refer to PDF file attached) to an end voltage of 1.85 volts, careful calculations being made of the exact number of ampere-hours delivered. On recharge the same number of ampere-hours are put back at the same current. A second discharge shall than be made to the same out-off voltage as before. The efficiency of the cell is then calculated as the ratio of the ampere-hour delivered during the second discharge to the ampere-hour put in on the charge.

11.10.1.1 Requirement

The ampere-hour efficiency when calculated as described in 11.10.1 shall be not less than 10 percent

11.10.2 Watt-Hour Efficiency

The watt-hour efficiency shall be calculated by multiplying the ampere-hour efficiency by the ratio of average discharge and recharge voltage. The values of discharge and recharges voltage shall be calculated from the log sheets for ampere-hour efficiency.

11.10.2.1 Requirement

The watt-hour efficiency when calculated as described in 11.10.2 shall be not less than 75 percent.

11.11 Test for Voltages Daring Discharge

The cell need not be discharged specially for this test. For the purposes of this test the voltages shall be obtained from the log sheets for capacity test in which the cell meets the rated capacity.

11.11.1 Requirement

The closed-circuit voltage of cell shall not be below the values given in co1 2.

Duration After Commencement of Discharge	Closed-circuit Voltage at Cell Terminals
(1)	(2)
a) Six minutes after putting on load	1.99 Min
b) 7 hours	1.92 Min
c) 10 hours	1.85Min

Commercialization rules

Notification / Declaration

What are the general rules in order to comercialize a product ?
Is there a notification / declaration to do ?

Trade license will have to be acquired which is governed by Municipality Act- which will location specific.
Notification is location specific.

Documents

What are the required documents ?

General documents will be required-in relation to establishment of business.

Delay

What is the delay to obtain the right to commercialize ?
Not Applicable

Display in store

Is there some specific rules on the way to display the product in store ?

Not Applicable

Display on internet

Is there some specific mentions to display the product on internet ?

Information to consumers

Do some specific information to have be given to consumers when the product is placed on the market ?

Not Applicable

Comments

Comments

Administration

What is the administration in charge of commercialization ? Name and address ?

Department of Legal Metrology which comes under the Department of Consumer Affairs.

Documents